Introduction to Operating System

WHAT IS OPERATING SYSTEM?

Operating system is an interface between user and the computer hardware. The hardware of the computer cannot understand the human readable language as it works on binaries i.e. 0's and 1's. Also it is very tough for humans to understand the binary language, in such case we need an interface which can translate human language to hardware and vice-versa for effective communication.

TYPES OF OPERATING SYSTEM:

- Single User - Single Tasking Operating System
- Single User - Multitasking Operating System
- Multi User - Multitasking Operating System

SINGLE USER - SINGLE TASKING OPERATING SYSTEM

In this type of operating system only one user can log into system and can perform only one task at a time.

E.g.: MS-DOS

SINGLE USER - MULTI TASKING OPERATING SYSTEM

This type of O/S supports only one user to log into the system but a user can perform multiple tasks at a time, browsing internet while playing songs etc.

E.g.: Windows 10, Window 11.

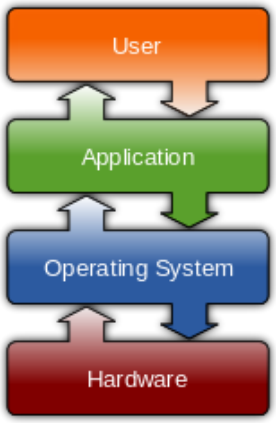
MULTI USER - MULTI TASKING OPERATING SYSTEM

This type of O/S provides multiple users to log into the system and also each user can perform various tasks at a time. In a broader term multiple users can logged in to system and share the resources of the system at the same time.

E.g.: UNIX, LINUX, Window Server OS. etc.

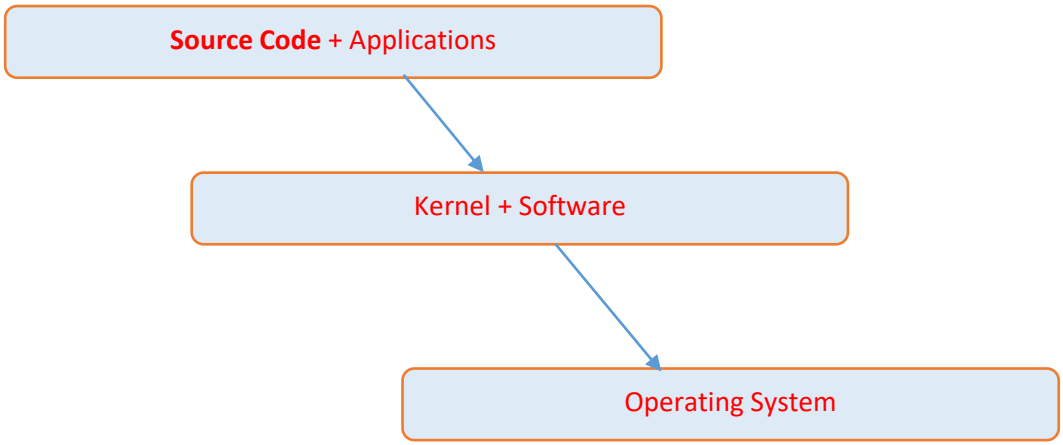
Function of an Operating System

- Process Management
- Memory Management
- Device Management
- Storage Management
- Application interface
- User interface
- Security
- Networking



Linux Origin

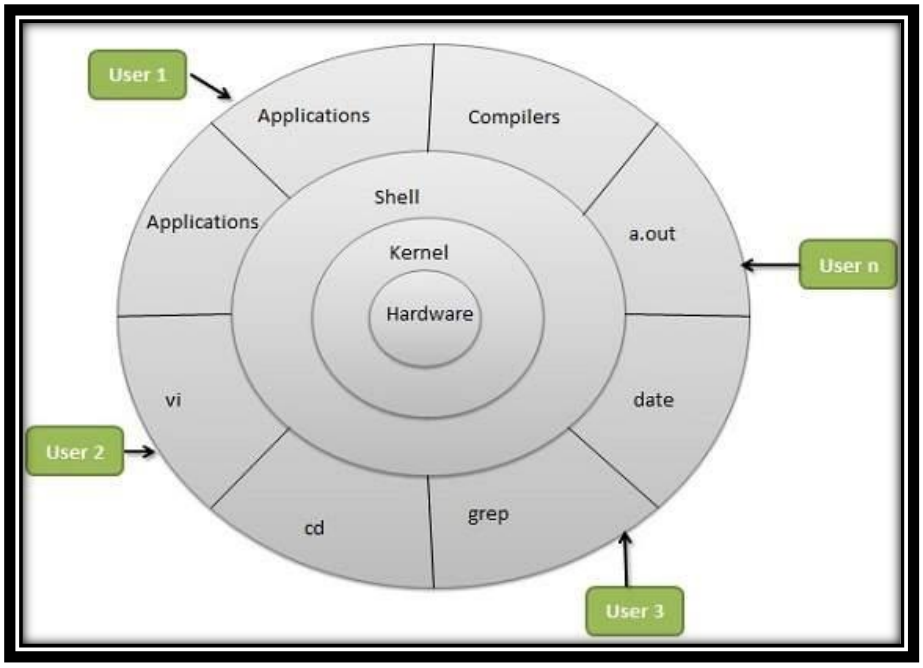
- **LINUS TORVALDS** – Finnish college student in 1991
- Linux was developed in 1991 by Linus Torvalds and a band of programmers who voluntarily developed the core program of the system, the kernel. That program was originally compatible for another operating system called Minix, but later development made it usable with GNU software.
- When Linux Kernel combined with GNU applications, complete free UNIX like OS was developed.



Features of Linux

- **Open Source** : Free software along with the source code and documentation
- **Multitasking** : Capable of running multiple applications and process at the same time.
- **Multi-user** : Allows multiple users to login and use the resources at the same time.
- **Portable** : Can be installed on all hardware architecture.
- **Scalable** : Same operating system can be used on a desktop to a super computer.
- **Reliable** : Large servers have been successfully being running without a single second of down time.
- **Secure** : Inbuilt firewall (iptables) and SELinux

Linux Architecture



The architecture of a Linux System consists of the following layers:

- **Hardware layer**: Hardware consists of all peripheral devices (RAM/ HDD/ CPU etc).
- **Kernel**: It is the core component of Operating System, interacts directly with hardware, and provides low level services to upper layer components.
- **Shell**: An interface to kernel, hiding complexity of kernel's functions from users. The shell takes commands from the user and executes kernel's functions.
- **Utilities:** Utility programs that provide the user most of the functionalities of an operating systems.

Installation of Linux OS

Flavors of Linux:

Novell	Suse Linux	1992
DEBIAN	Debian	1993
RedHat	RHE/RHEL/fedora	1995
CentOS	CentOS	2004
Canonical Ltd	Ubuntu	2004

Installation of RHEL:

There are primarily three steps in installing any Operating System, which are as below:

1. Pre-installation Task
2. Installation Task
3. Post Installation Task

Preinstallation Task:

Minimum and Recommended Requirements to install RHEL 8 & RHEL 9 are:

Hardware	Minimum Requirement for RHEL8-64BIT	Recommended Requirement for RHEL8-64BIT	Minimum Requirement for RHEL9-64BIT	Recommended Requirement for RHEL9-64BIT
PROCESSOR	1.1 GHz 64-bit processor (x86_64)	2 GHz or faster multi-core processor	64-bit processor supporting x86-64-v2 microarchi tecture	2 GHz or faster multi-core processor
MOTHER BOARD	VT ENABLED	VT ENABLED	VT ENABLED	VT ENABLED
RAM	1.5 GB for text-based installation; 2 GB for graphical installation	4GB or more	2GB or more	4GB or more
HARD DISK	10 GB for minimal installation	20 GB or more, depending on selected packages	10 GB for minimal installation	20 GB or more, depending on selected packages

Minimum Partition creation and sizes for basic installation

Partition Name	Size For 64 Bit
/ (root)	20 GB - Core OS files, libraries, binaries, logs, etc.
/boot	1 GB Contains bootloader and kernel files (non-LVM, ext4 recommended).
SWAP	Twice of RAM (Varies depend on size of RAM) RAM Size — Recommended SWAP ≤2GB — 2 × RAM 2 – 8GB — = RAM > 8GB — 4 – 8 GB
Other non-mandatory*** /home /var /opt	 For user files and directories For Logs, mail, spool files, etc For third party software installations.

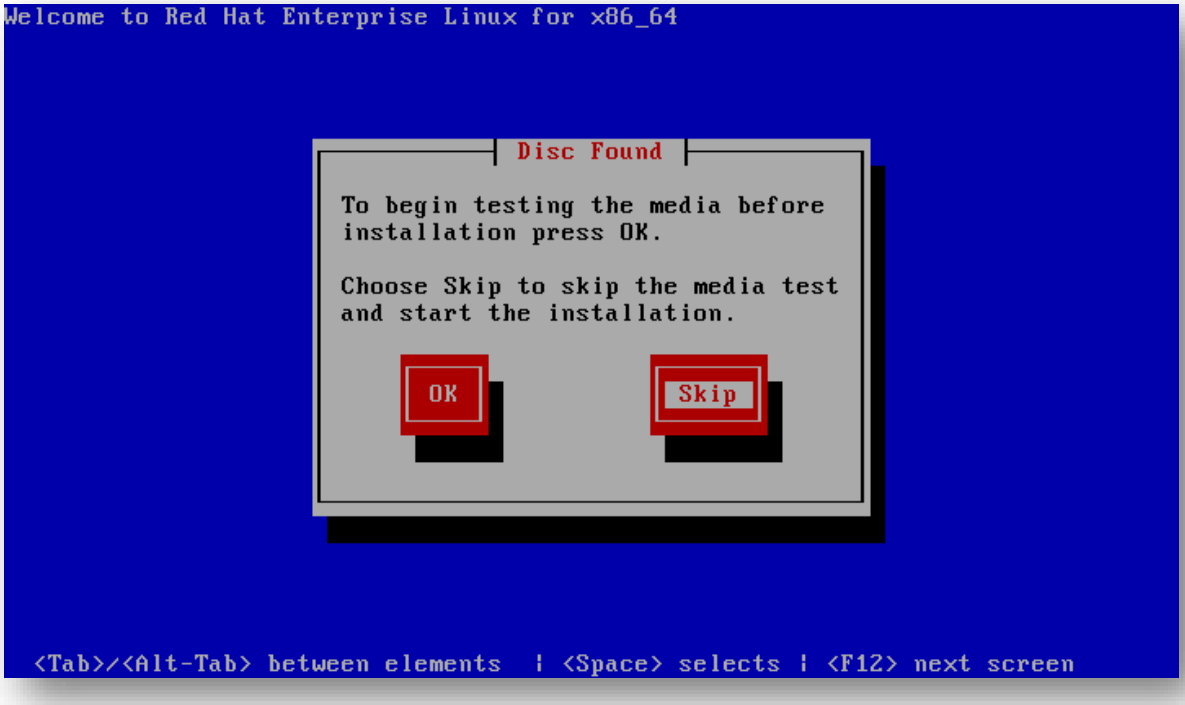
*Table 1: ***Other non-mandatory partitions:*

Partition	Purpose	Why Separate? (Key Points)
/home	User files & profiles	Protect / from user data, safer OS upgrades
/var	Logs, mail, DBs	Prevent log overflow, enhance I/O, isolate critical services
/opt	3rd-party apps	Isolate custom software, security hardening, easy management

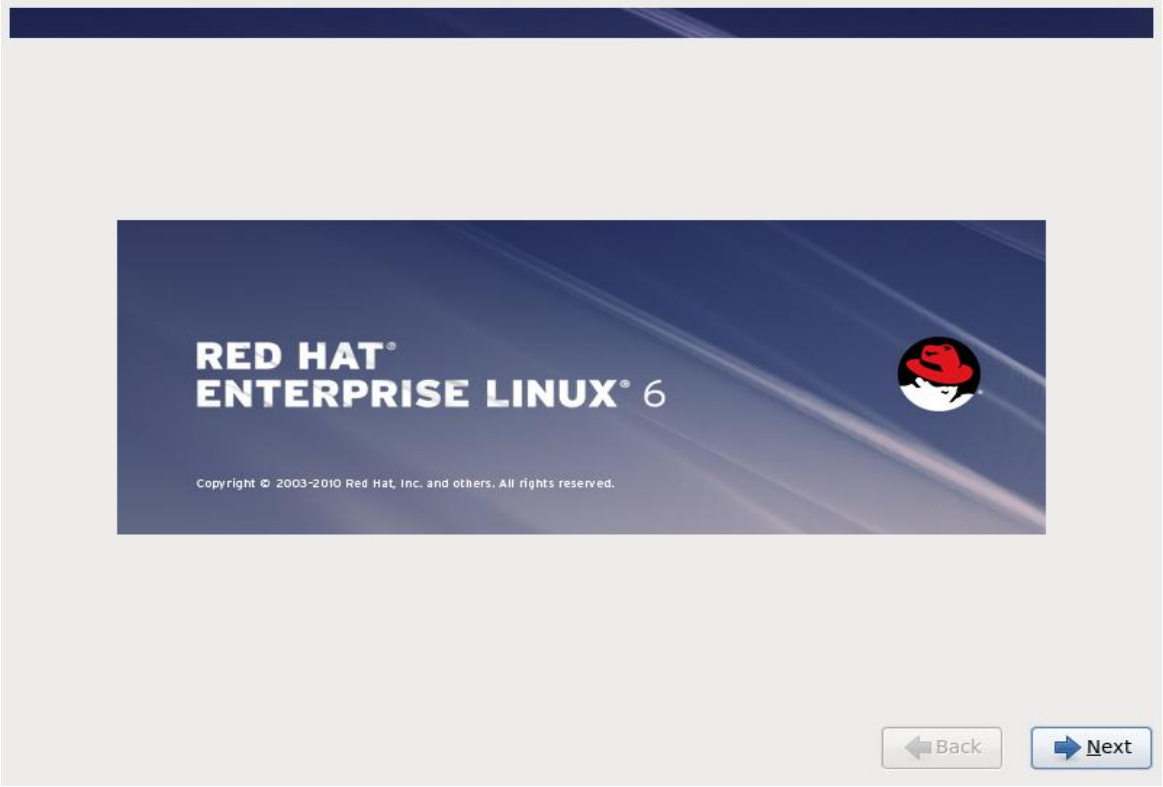
Installing RHEL6 with above specification

- Enter into BIOS setting and make CD/DVD Drive as first boot device
- Make sure that VT (Virtual Technology) is enabled for RHEL6-64-bit systems
- Insert the RHEL 6 CD/DVD into CD/DVD drive and boot the system
- If booted from CD/DVD Rom the following screen will be displayed

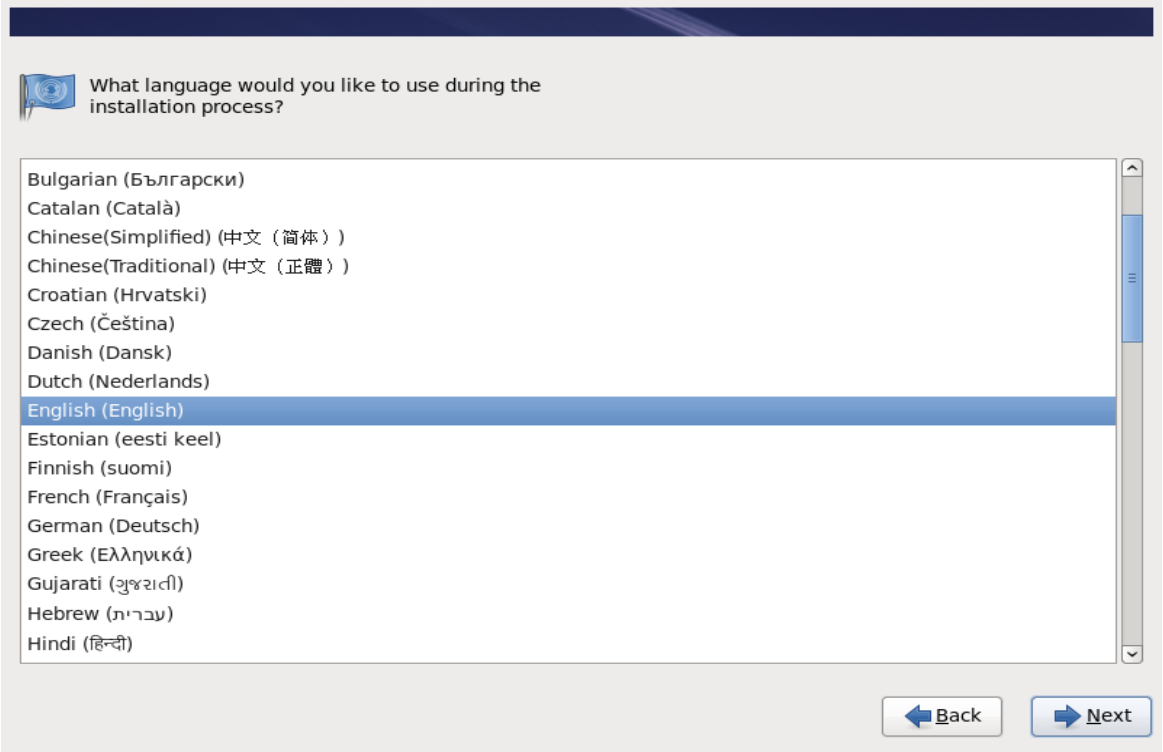
- Move the cursor to **Install or upgrade an existing system** and press **Enter**



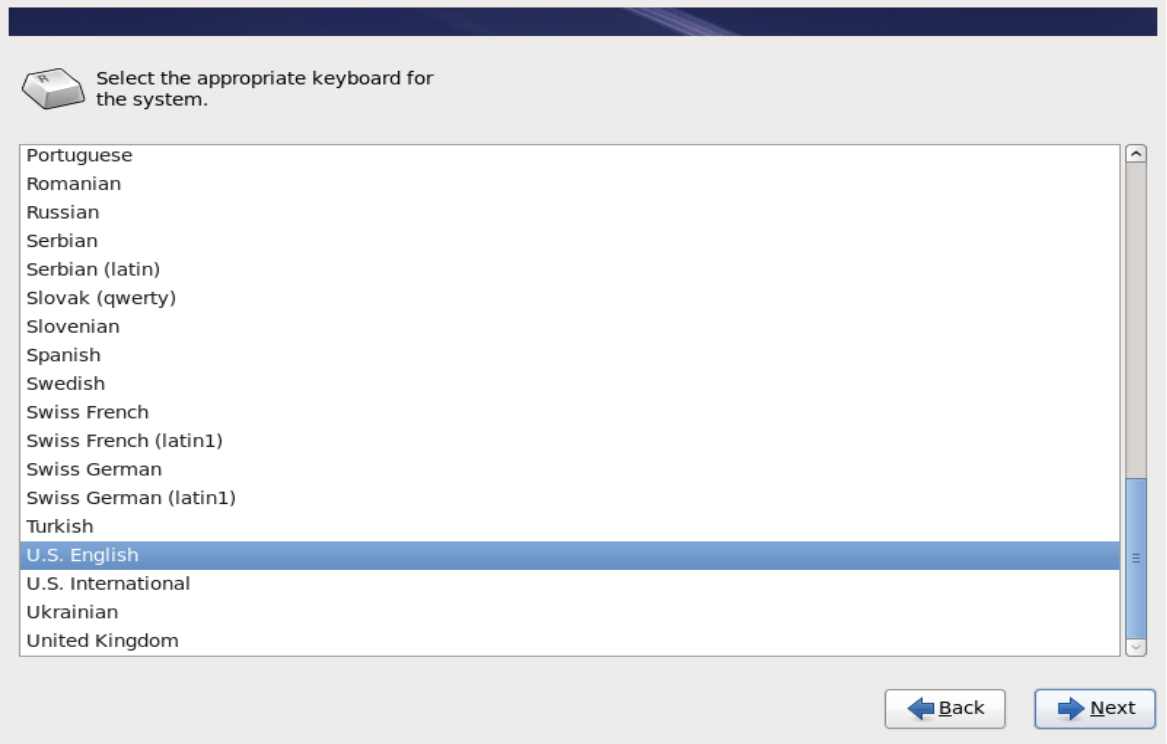
- To test the media select **OK**, to skip the testing move cursor to **Skip** and press enter



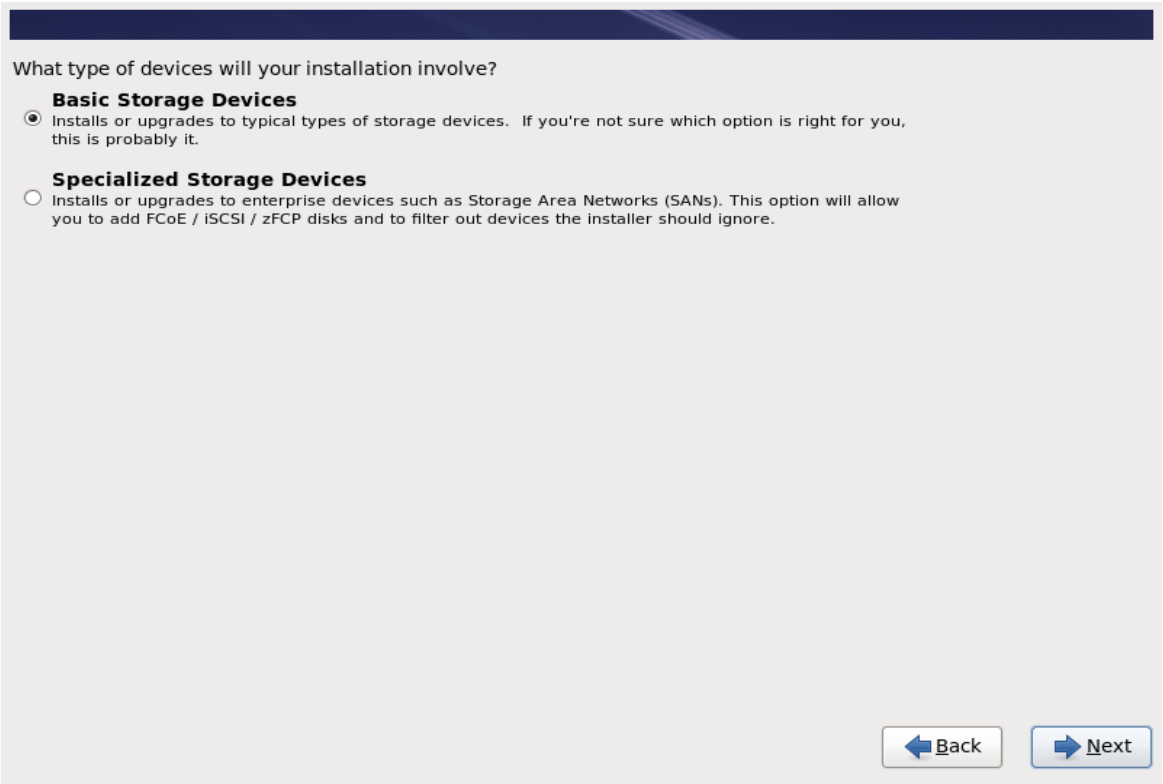
- Click on **Next** button to move forward



- Select your desired language, usually **English.** Click **Next** to continue

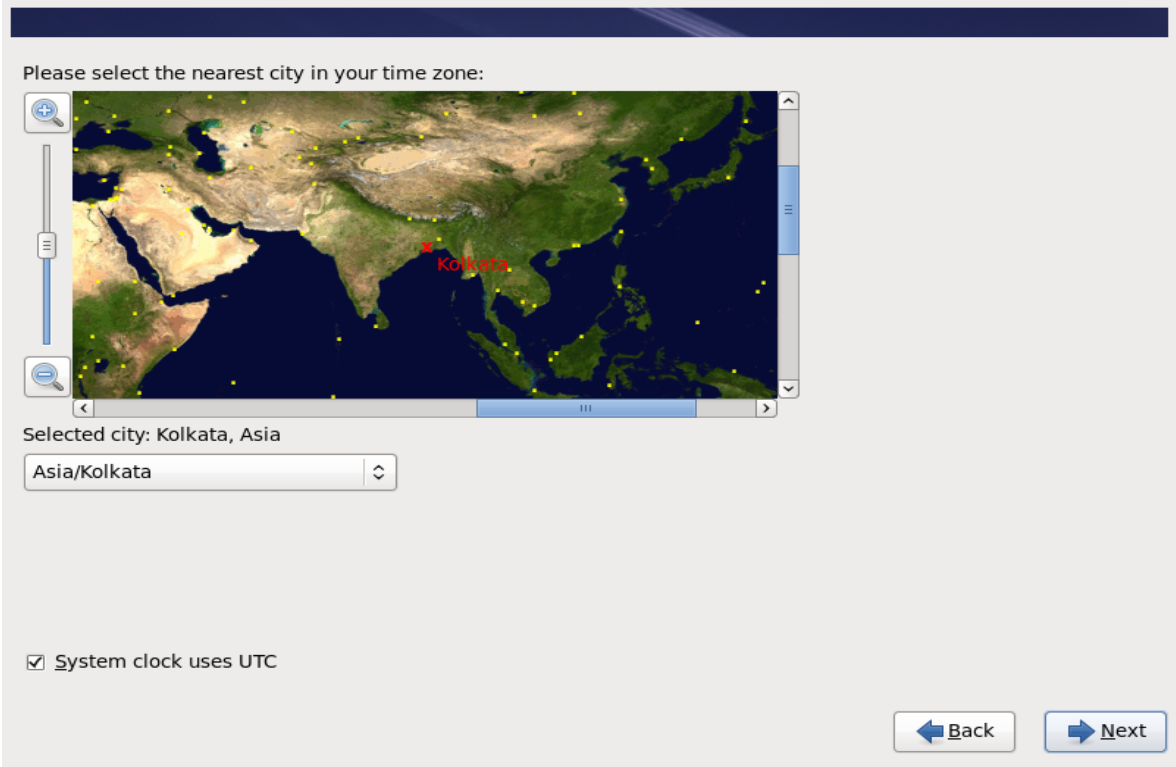


❖ Select the keyboard type as required usually **U.S English**, click **Next** to continue

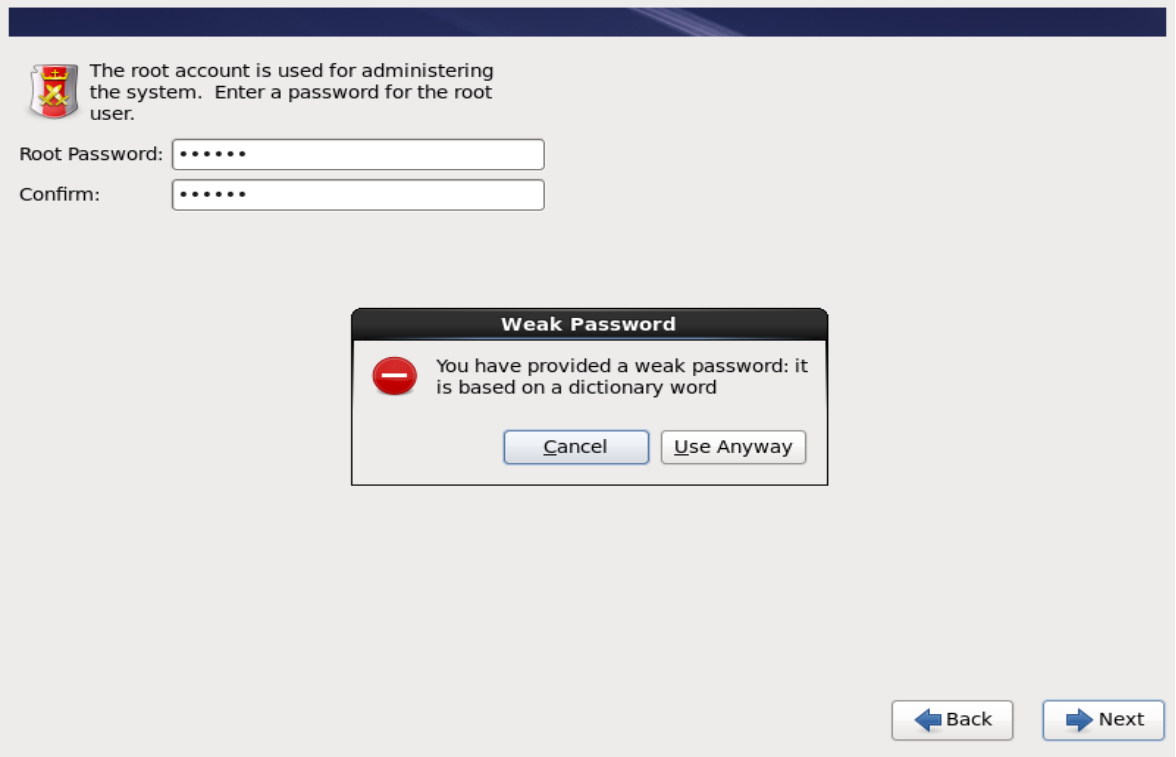


❖ Select the type of storage for the Computer. Click **Next** to continue

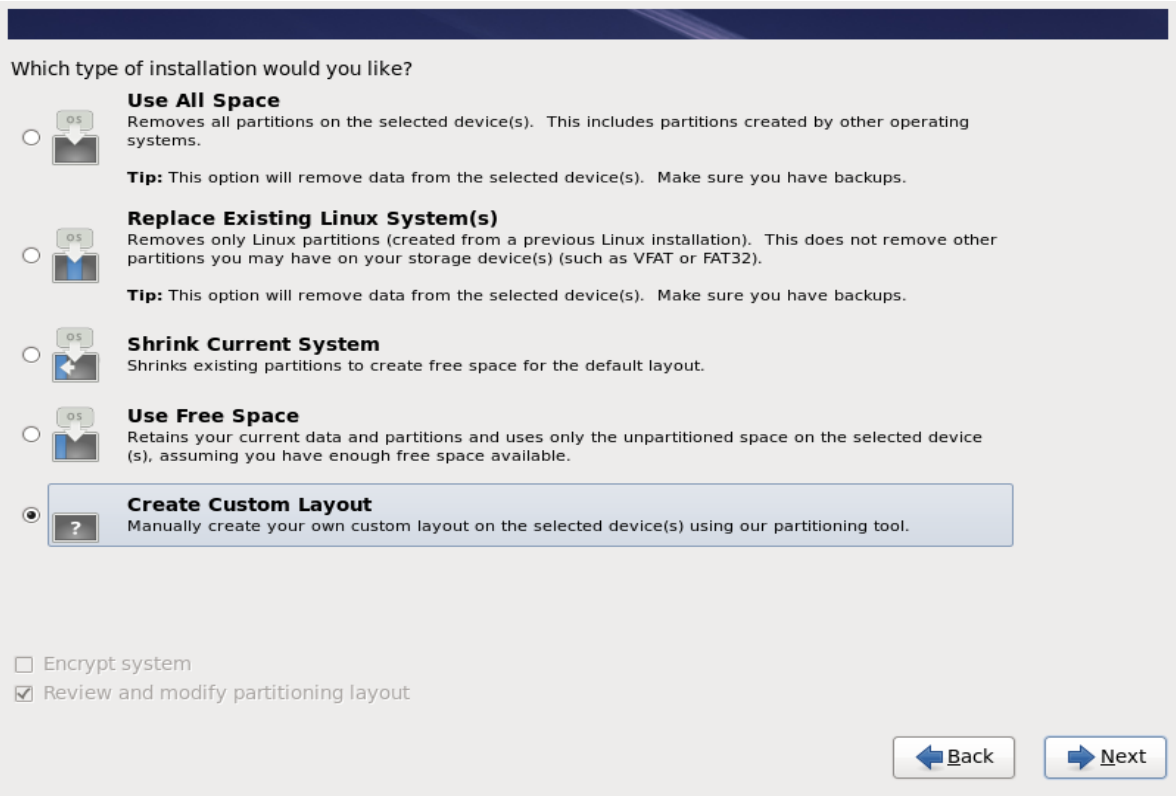
❖ Assign a hostname to the system, if wish to give ip address click on Configure Network, else Click Next to Continue



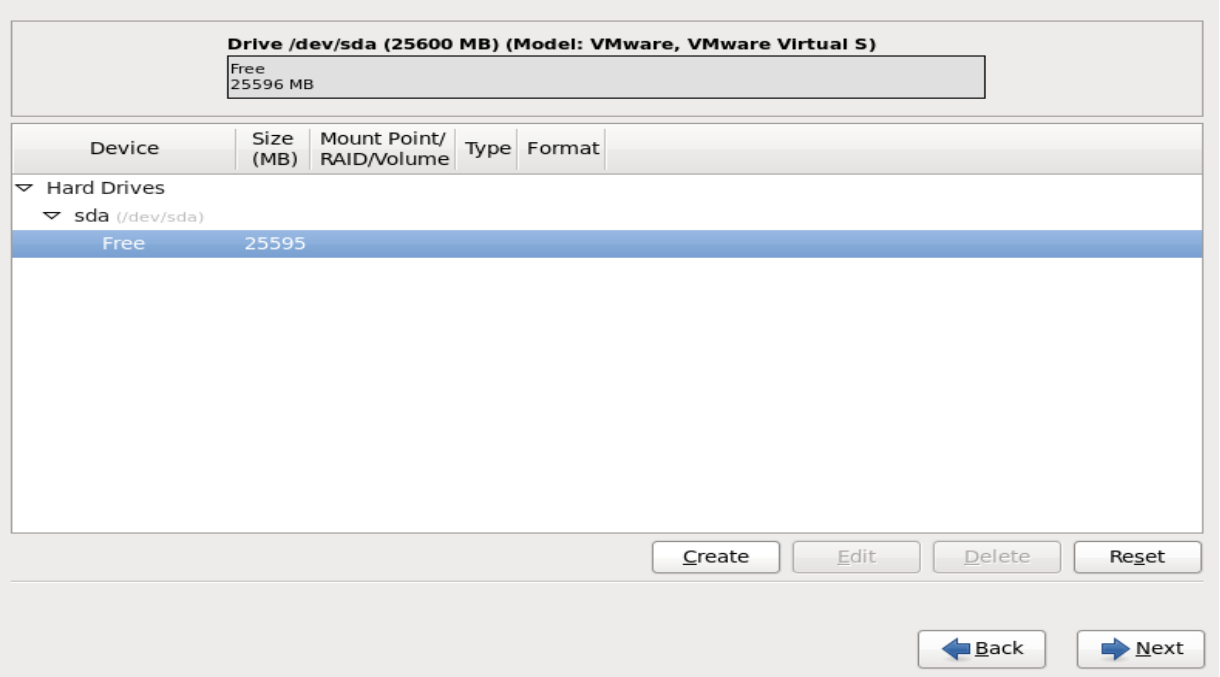
❖ Select the nearest city in your Time Zone and Click on **Next** to continue



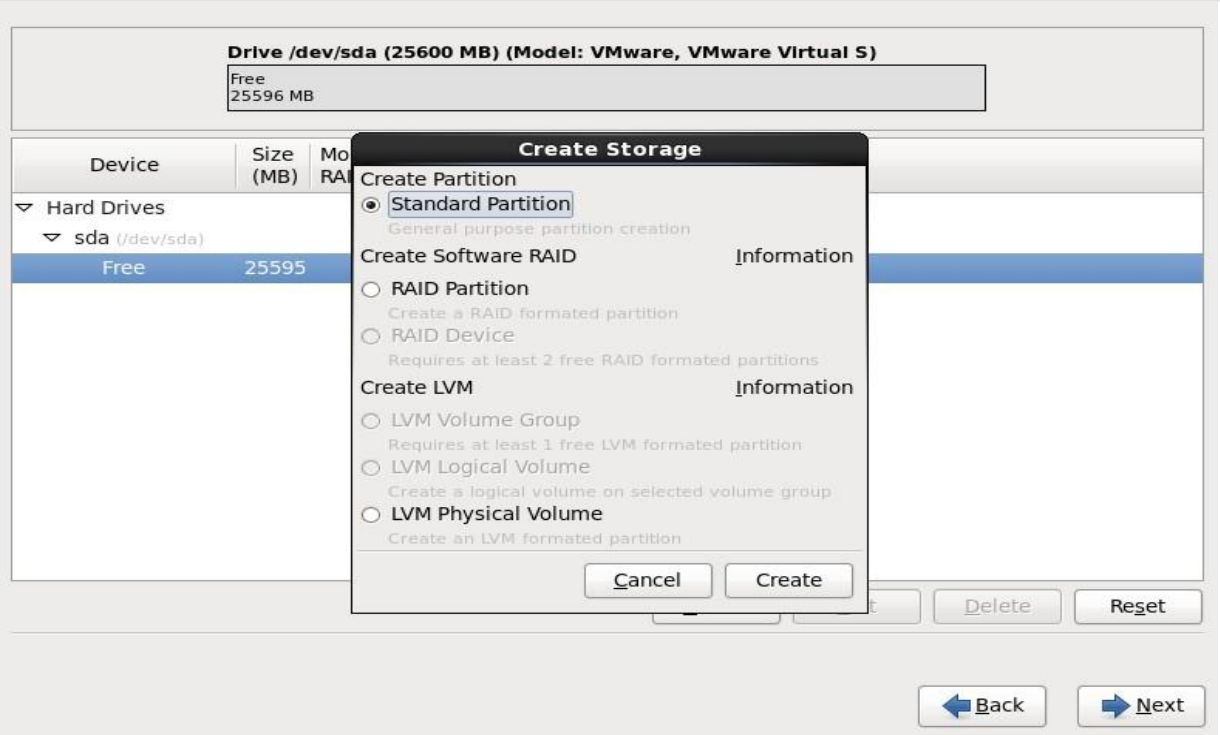
❖ Assign some complex password for **root**, then click on **Next** to continue



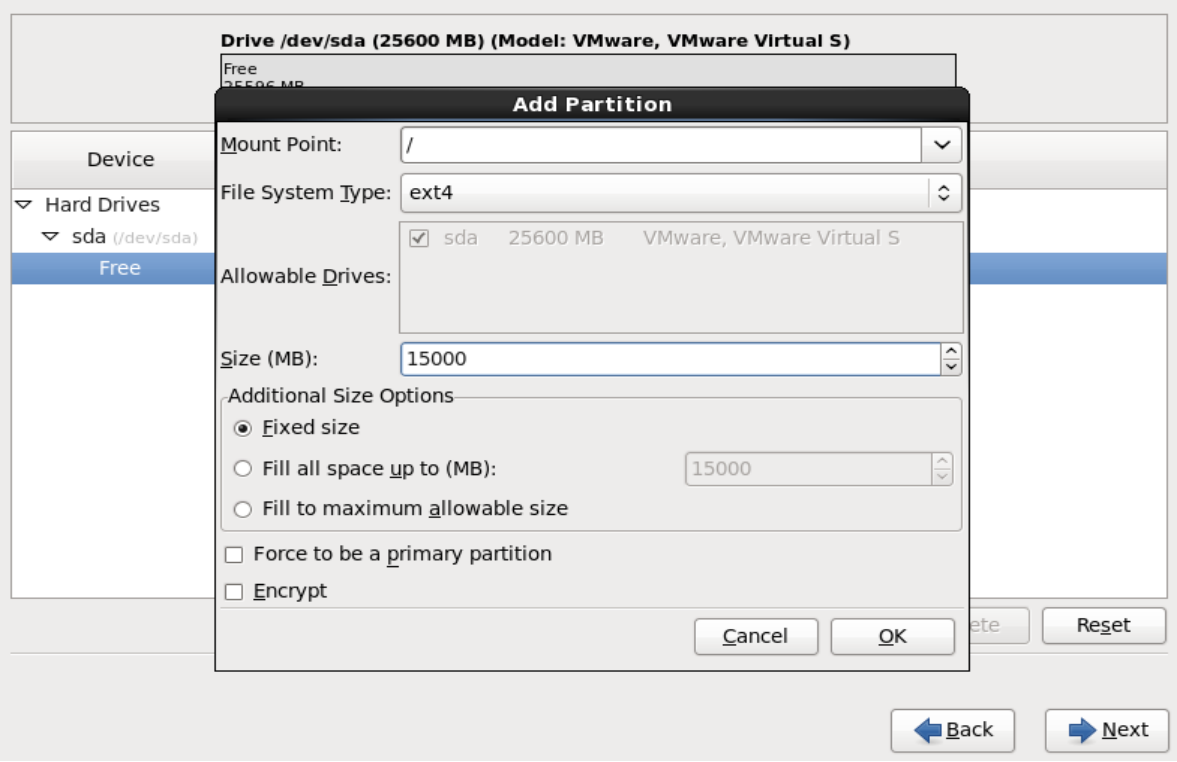
❖ Select the type of partitioning you want, to create your own partitions with custom sizes, select **Create Custom Layout** and click on **Next** to continue



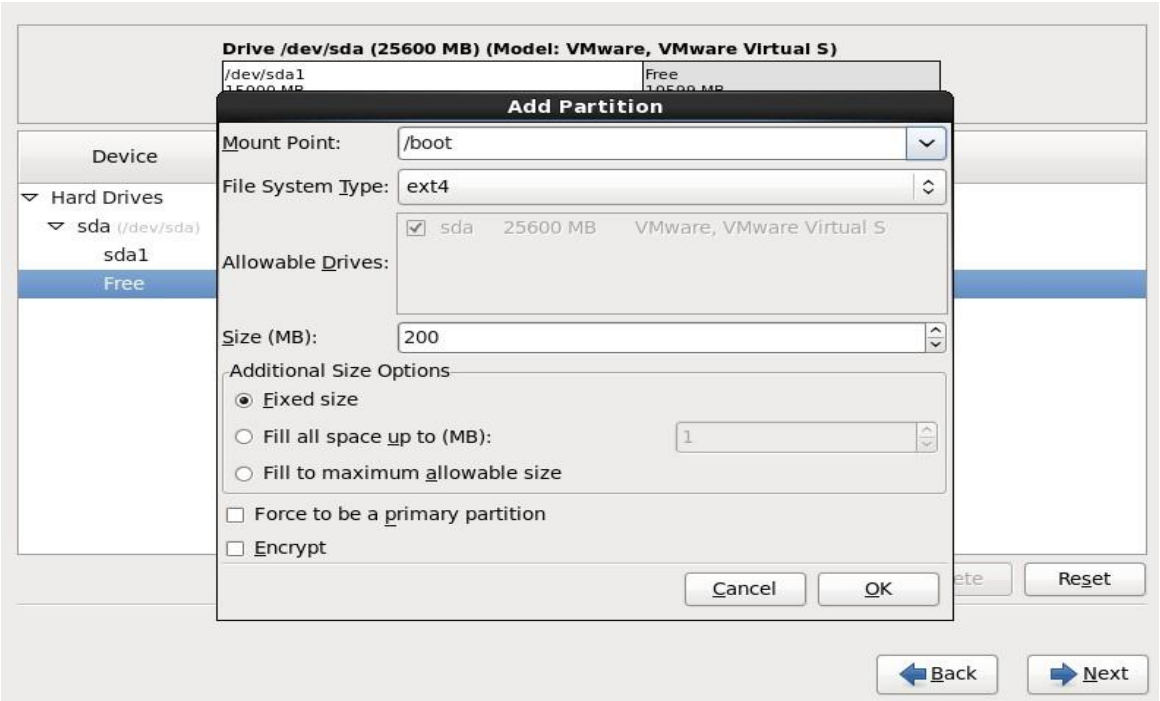
❖ Click on the **Free** space, then click on **Create** to create your own partitions



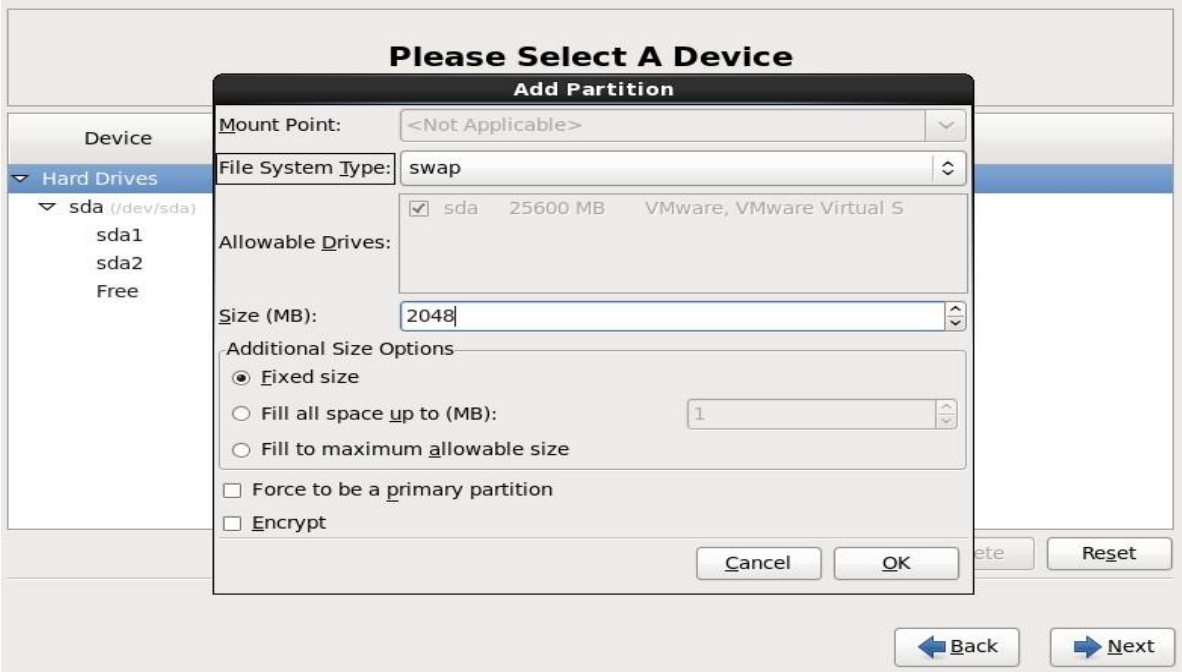
❖ Check the box beside **Standard Partition,** Click on **Create** to continue.



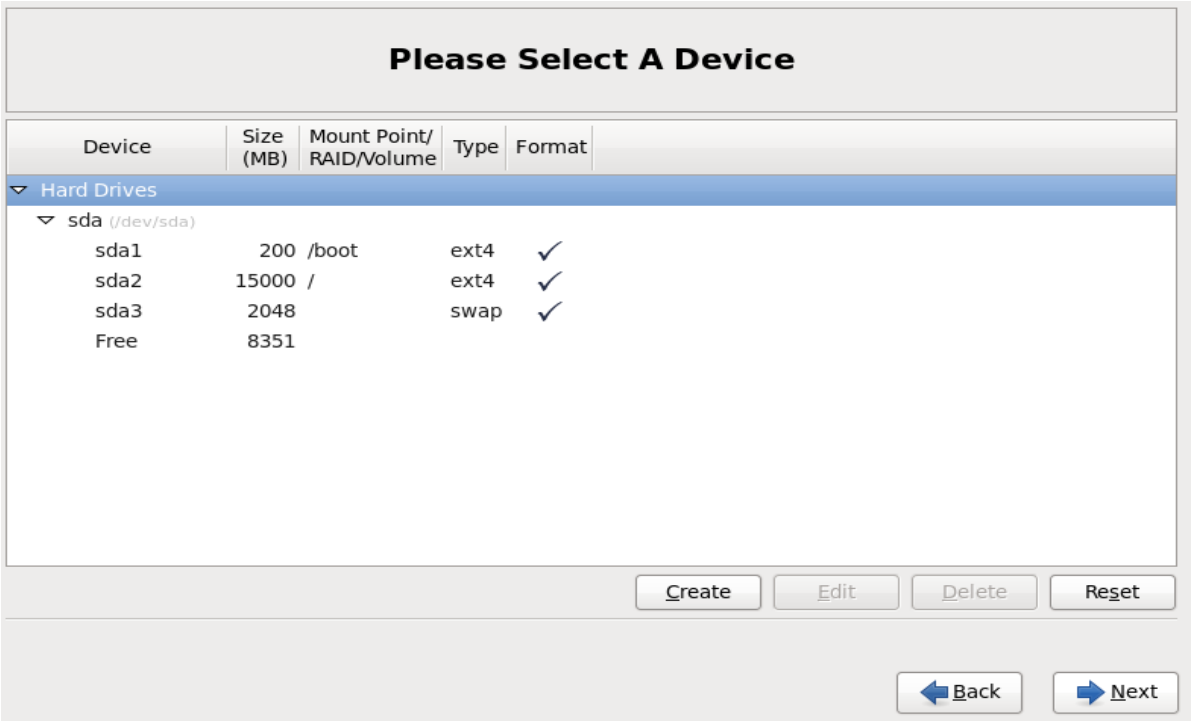
- Select **/** from **Mount Point** Box, give the size you wish for it and click on **OK** to create it.
- Select the **Free** space again and click on **Create** to create another partition. Also Check the box beside **Standard Partition**, Click on **Create** to continue



- Select **/boot** from **Mount Point** Box, give the size 2OO MB for it and click on **OK** to create it.
- Repeat the same steps and create swap space



❖ This time select **swap** from **File System Type**, give the size required and click on **OK**



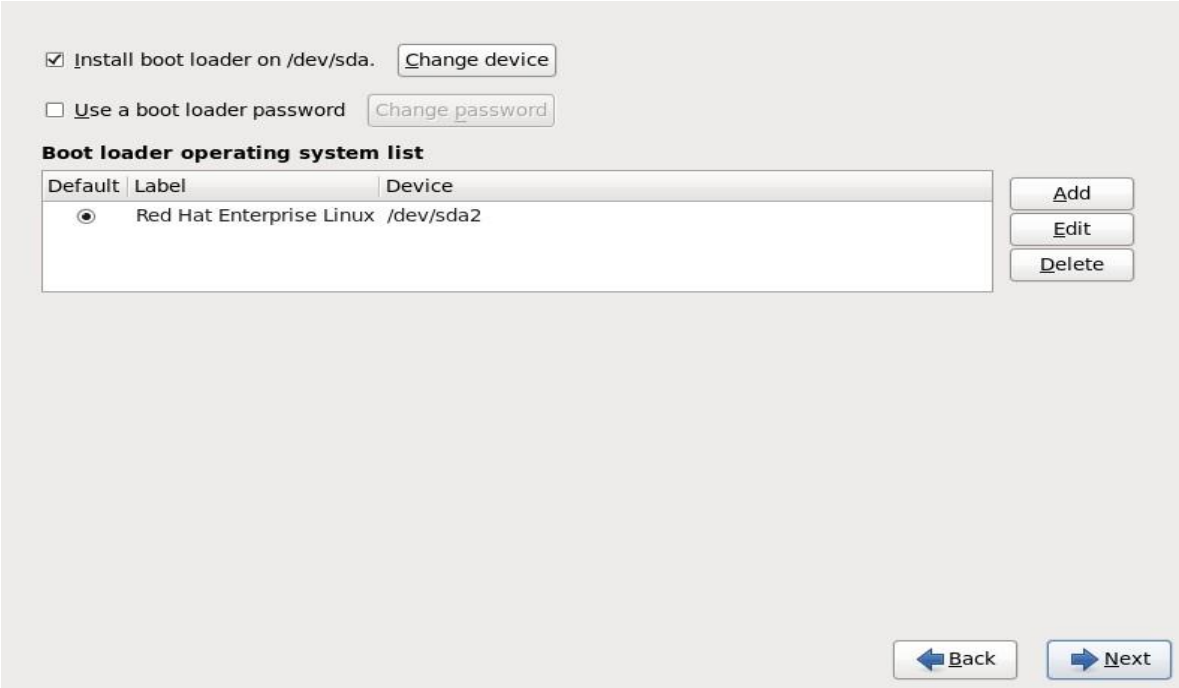
❖ Verify the partition and click on **Next** to continue with it.

❖

❖ Click on Format to **Format** the partition and continue with it.

❖ Click on **Write changes to disk** to continue, if wish make changes click on **Go back**.

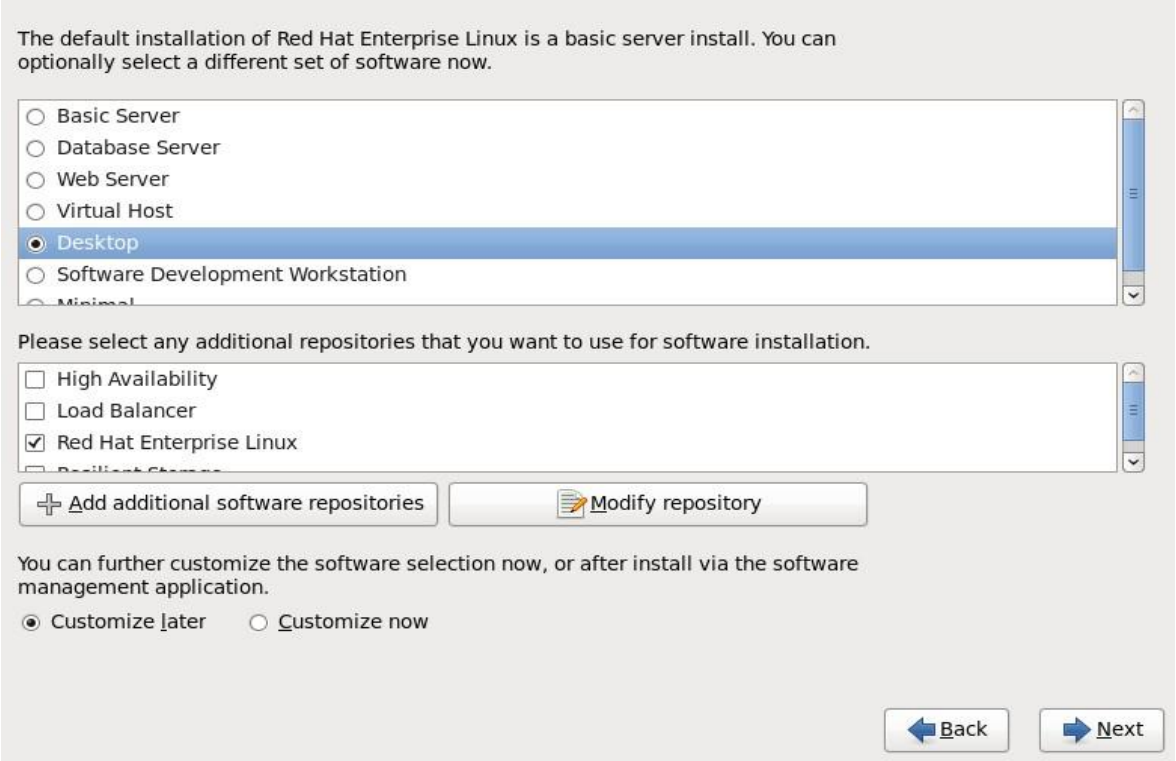
❖



❖ To change the name of boot loader select **Edit** and assign new name to it.

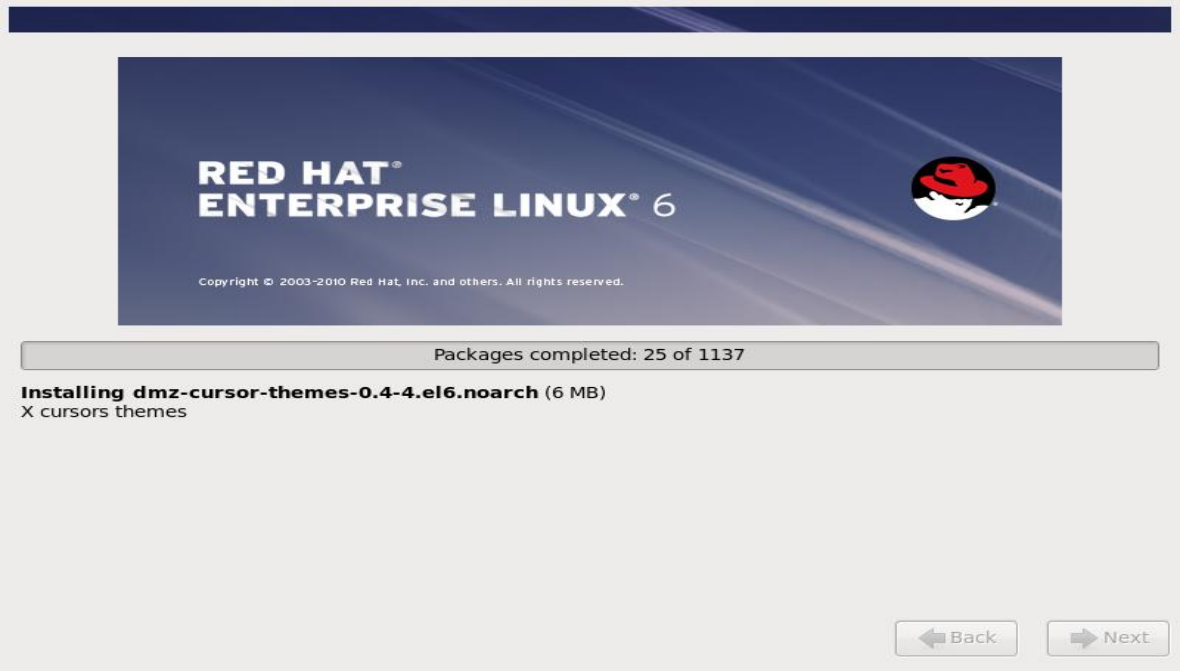
❖ To assign password to boot loader check the box beside **Use boot loader password** and assign a password to it.

❖ To keep all as default, just click on **Next** button to continue.

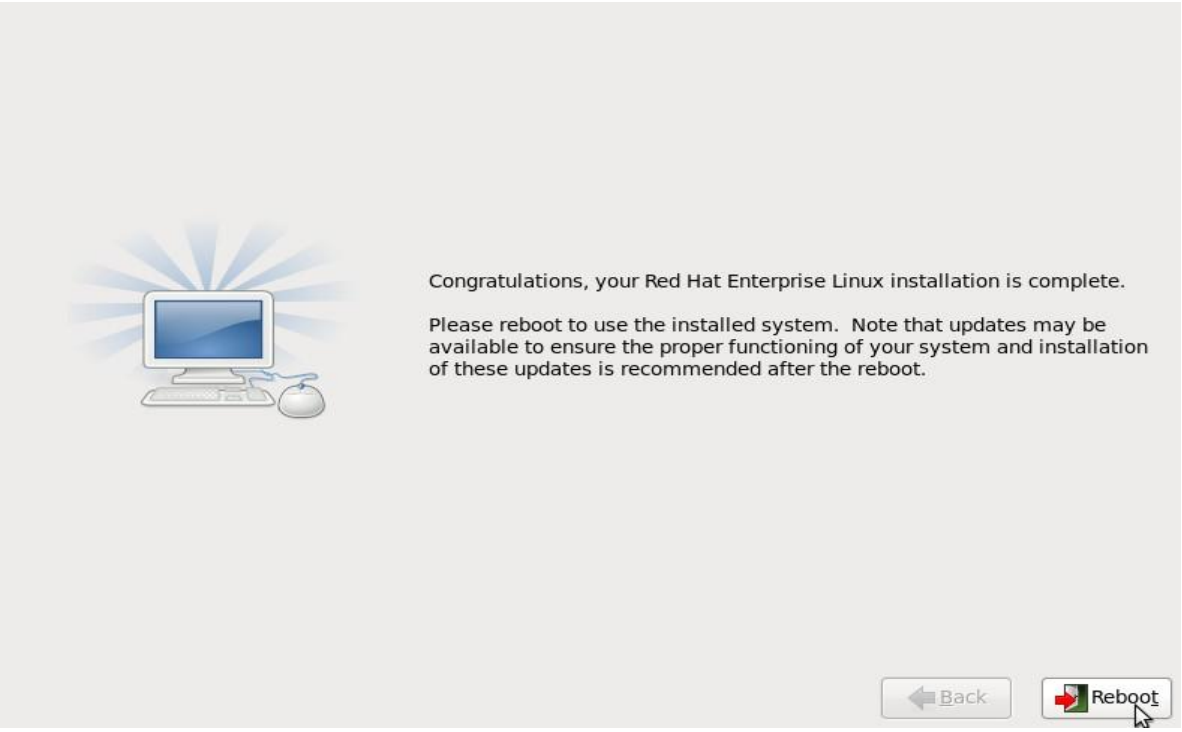


❖ Select **Desktop** to have a graphical environment in RHEL6.

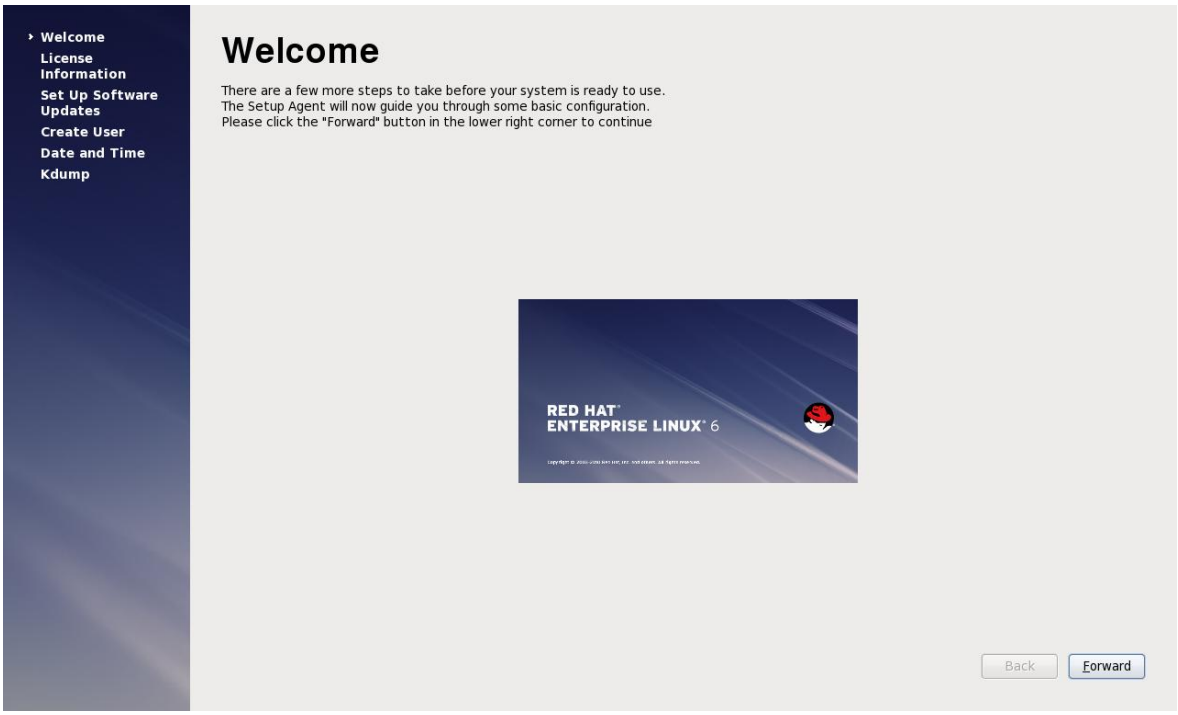
❖ Check **Customize later** to install additional software later. Click on **Next** to continue



Now sit back and relax until the installation is completed

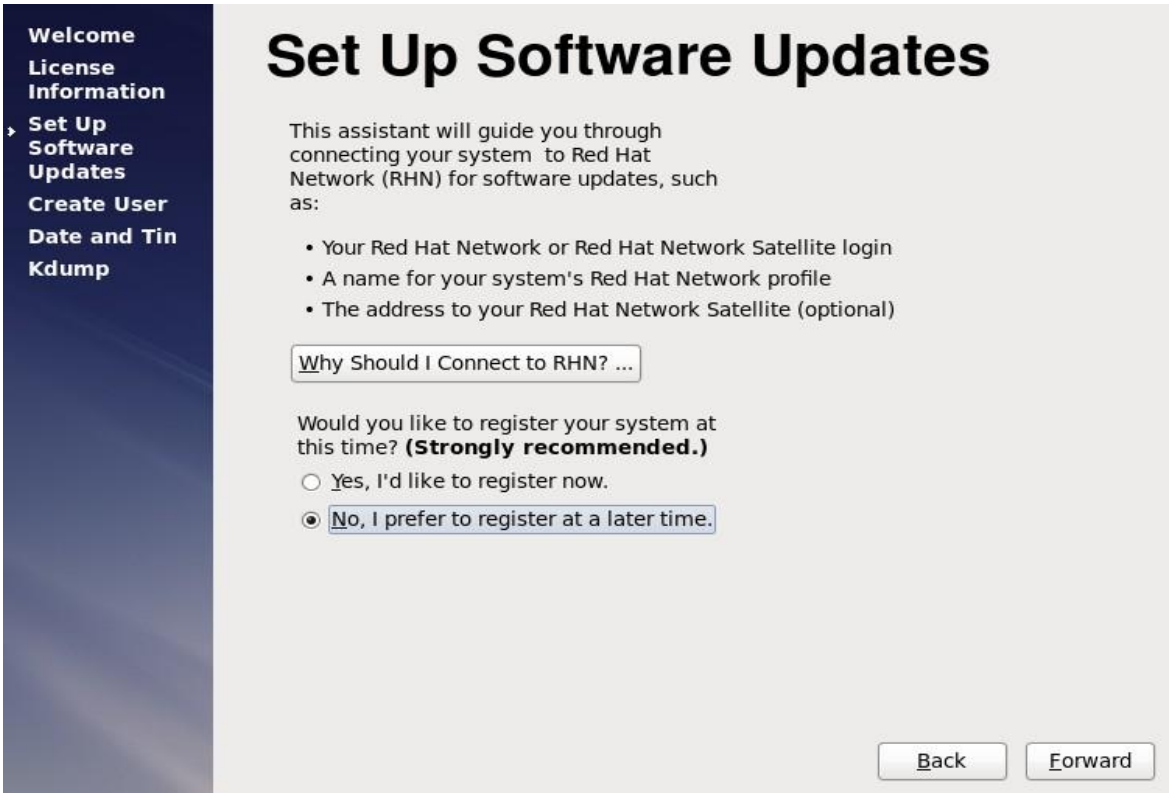


- When above prompt is displayed, remove the CD/DVD from the drive and click on **Reboot** to reboot the system.

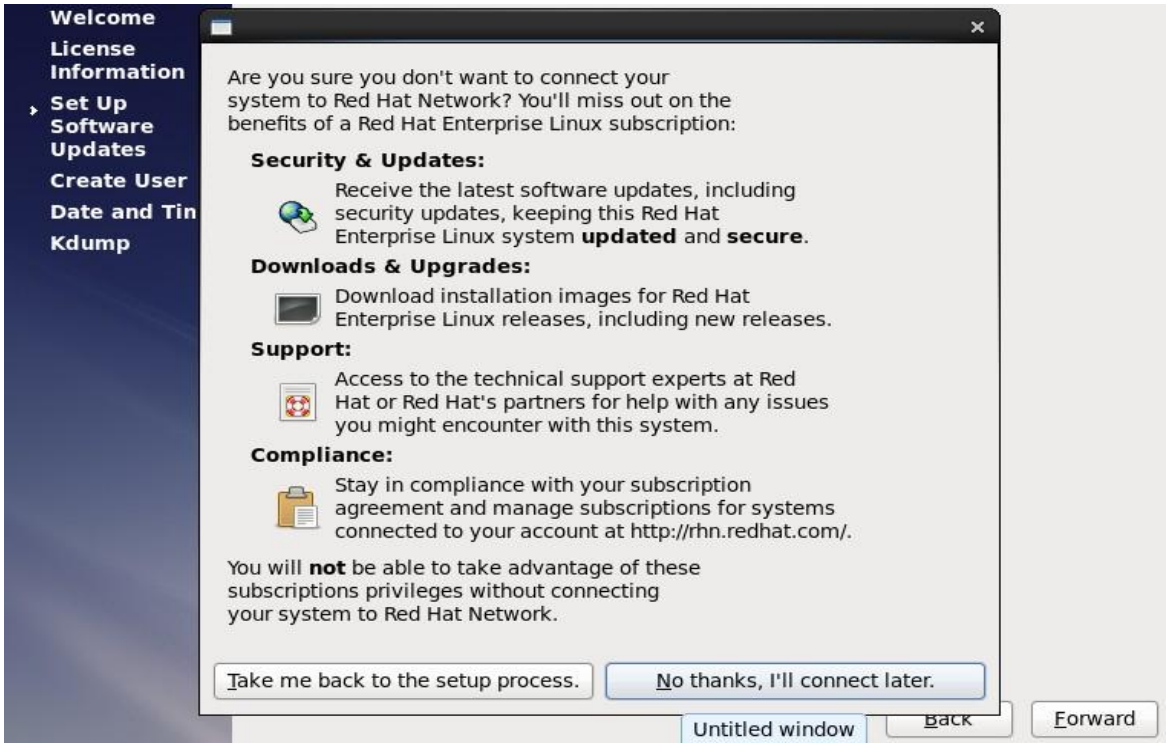


- Click **on Forward** to move to next step.

❖ **Accept** the license agreement and click on **Forward** to continue

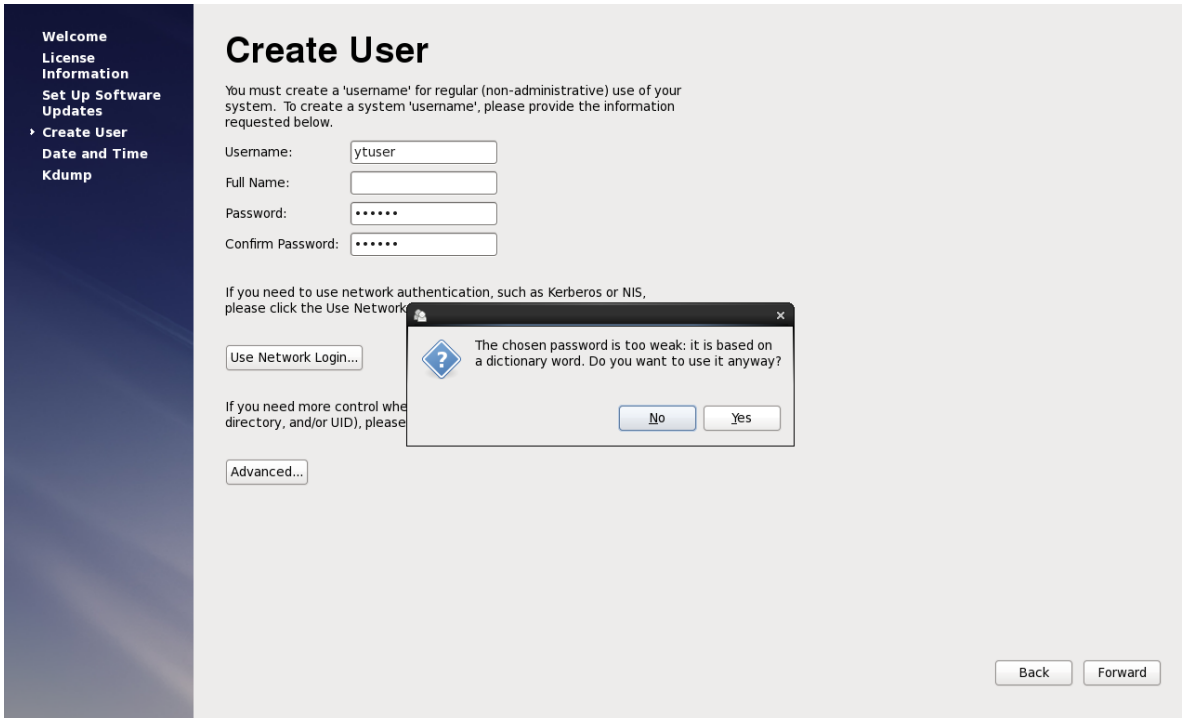


❖ Check **No, I prefer to register at a later time.** To skip the registration and click on **Forward**.

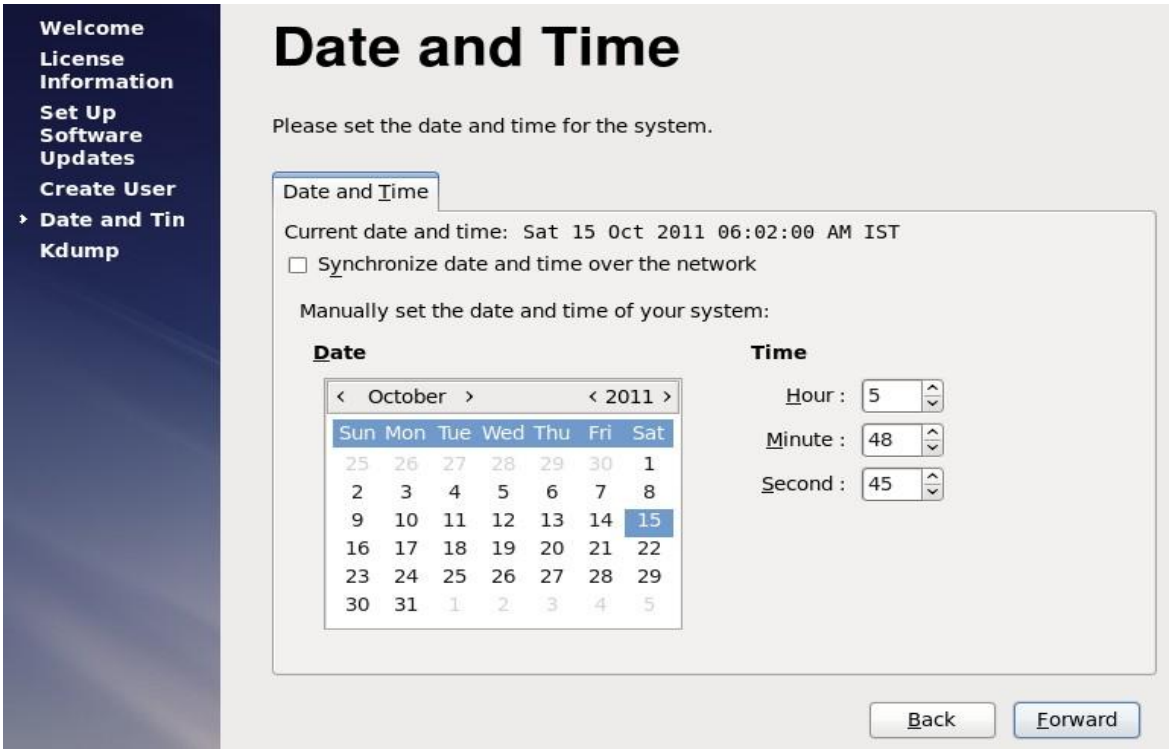


❖ Click on No Thanks, to Move to Next Step

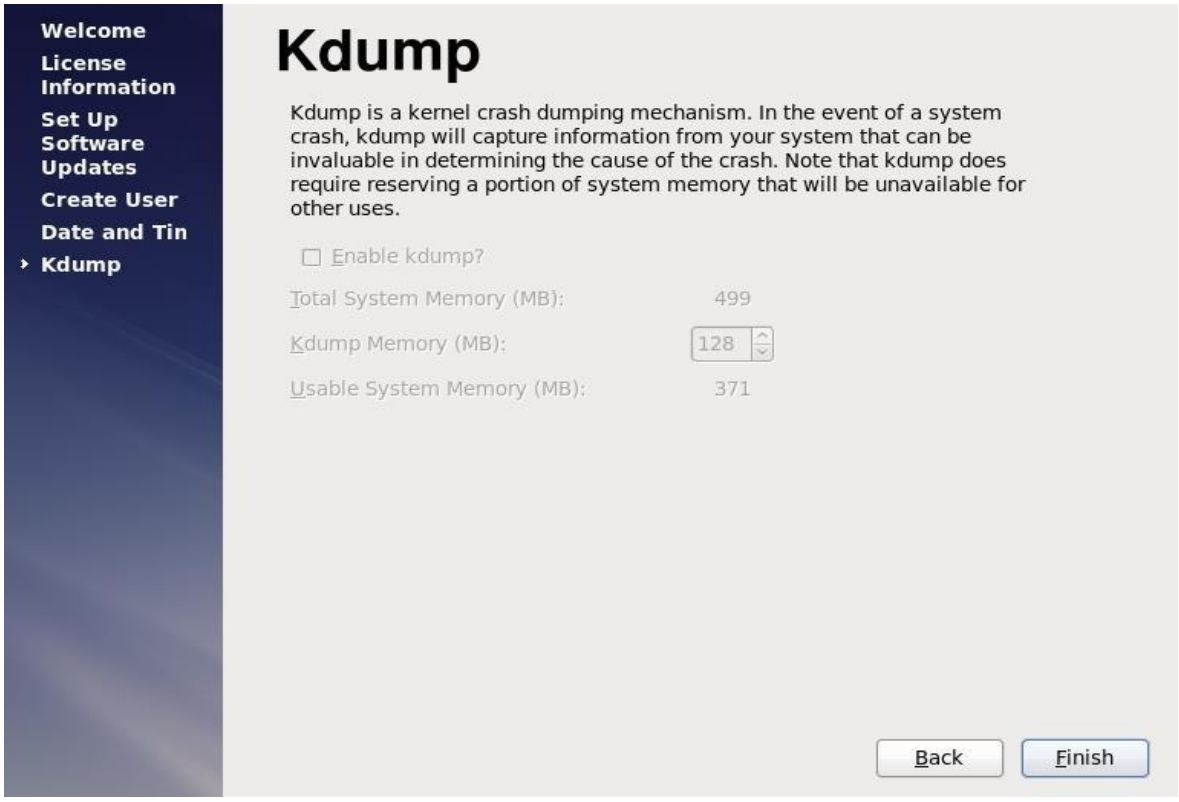
❖ Click on **Forward** to continue



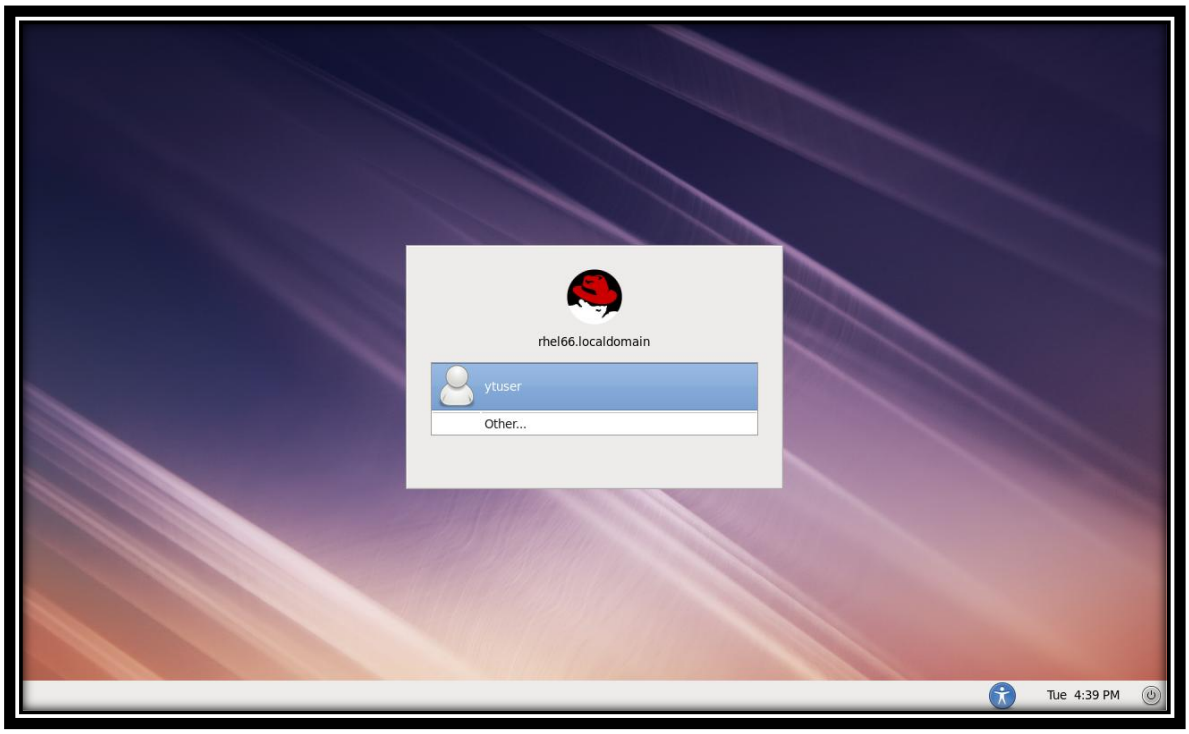
❖ Give a name to create a user and assign it a password. Click on **Forward** to continue.



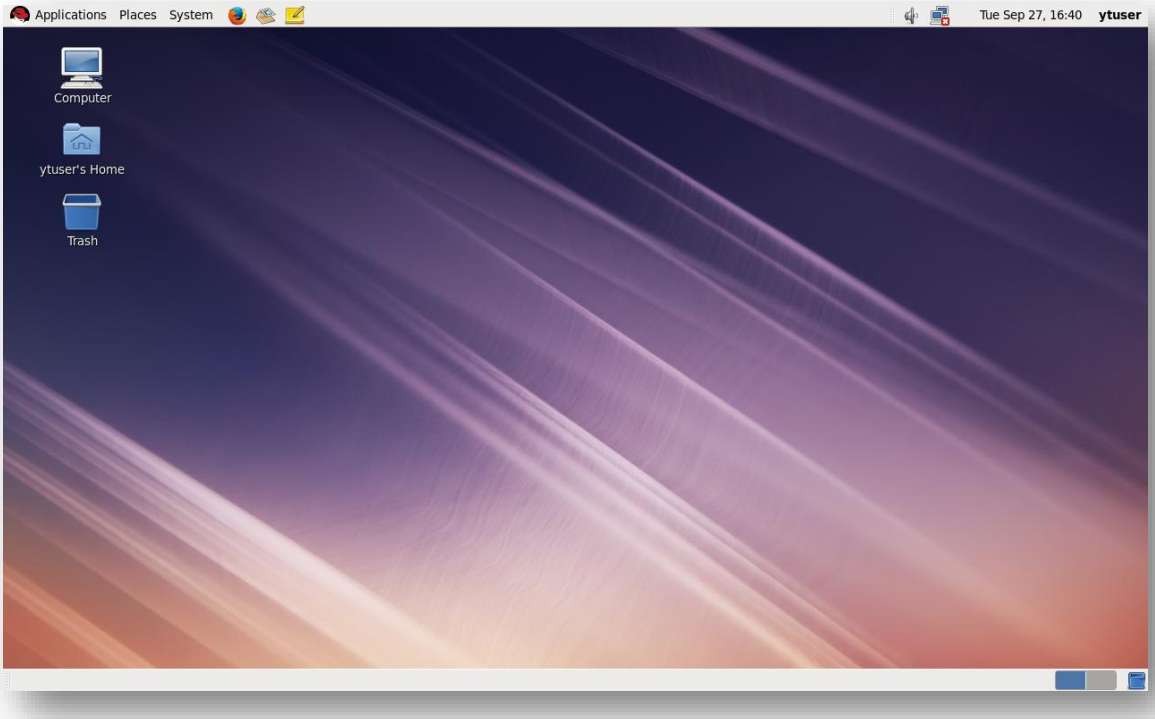
❖ Set the date and time and click on **Forward** to continue



- Click on **Finish** and congratulation your installation is now completed



- Login As an **ytuser OR root** user and explore the Linux World!!!


Applications
Places
System
Tue Sep 27, 16:40
ytuser
Computer
ytuser's Home
Trash

File System Hierarchy

All files on a Linux system are stored on file systems which are organized into a single inverted tree of directories, known as a file system hierarchy.

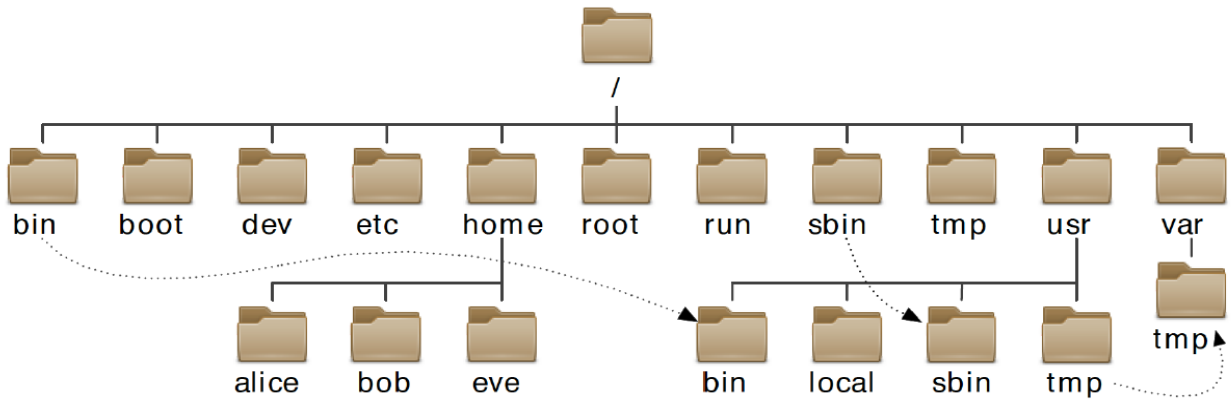


Figure 1: Linux file system hierarchy

File system directory contain or store different types of data as below:

- **Static** is a content that remains unchanged until explicitly edited or reconfigured. It is installed with operating system.
  - /bin, /sbin, /lib, /lib64, /usr, /etc.
- **Dynamic** or **Variable** is a content typically modified or appended by active processes.
  - /var, /tmp, /var/tmp
- **Persistent** is a content, particularly configuration settings that remain after a reboot.
  - /home, /etc, /var
- **Runtime** is a process or system specific content or attributes cleared during reboot.
  - /run, /proc, /sys, /dev

/ This is the top-level directory
It is parent directory for all other directories
It is called as ROOT directory
It is represented by forward slash (/)
C:\ of windows

/root It is the home directory for root user (super user)

It provides working environment for root user.
C:\Documents and Settings\Administrator

/home It is the home directory for all normal users (non-admin users)
It provides working environment for all the normal users.
c:\Documents and Settings\username

/boot It contains bootable files for Linux
Like vmlinuz (kernel)..... ntoskrnl
Initrd (INITial Ram Disk) and
GRUB (GRand Unified Boot loader).... boot.ini, ntldr

/etc It contains all configuration (settings) files (
Like /etc/passwd.... User info
/etc/resolv.conf... Preferred DNS
/etc/dhcpd.conf.... DHCP server
C:\windows\system32\dirvers\

/usr It's the default directory where software will be installed
(UNIX Sharable Resources)
c:\program files

/opt It is the optional directory for /usr
To install third party softwares
c:\program files

/bin It contains ordinary set of commands used by all users (non-admin)

(Binary files)

/sbin It contains commands used by only Super User (root) (Admin)
(Super user's binary files)

/dev it contains the device files
Like /dev/hda ... for hard disk
/dev/cd rom ... for cd rom
Similar to device manager of windows

/proc It contain ongoing process files
Its files contain useful information used by OS
Like /proc/meminfo... information of RAM/SWAP
/proc/cpuinfo... information of CPU
Like Task Manager of windows

/var It contains variable data like mails, log files

/tmp	It provides temp space for software/apps to keep its data for short period time while they are running.
/mnt	It is default mount point for any partition (non-removable media)
/media	It's the mount point for removable media like CD-ROM, pen drive
/lib	it contains library files which are used by OS It is similar to dll files of windows, Library files in Linux are SO (shared object) files